

Resistor Sorter and Retriever System

The Resistor Sorter and Retriever System is an automated embedded system designed to accurately identify, classify, and store resistors based on their resistance values. The system integrates sensing, processing, and actuation mechanisms to reduce manual effort and improve efficiency in handling electronic components. It is particularly useful in laboratories, workshops, and educational environments where large quantities of resistors need to be organized and accessed quickly.

At the core of the system is the ESP32 microcontroller, which acts as the main control unit. It coordinates all operations including data acquisition, processing, and control of motors and sensors. The identification of resistor values is achieved using a Wheatstone bridge circuit, which converts the unknown resistance into a measurable voltage difference. This analog signal is then fed into the ADS1115 analog-to-digital converter, which provides high-resolution digital data to the ESP32. Using calibrated calculations, the microcontroller determines the exact resistance value and categorizes the resistor into predefined ranges.

To ensure precise positioning of the sorting mechanism, a NEMA 17 stepper motor is employed. The motor is controlled using a TB6600 stepper motor driver, which provides accurate current control and smooth operation. A homing mechanism using the A3144 Hall-effect sensor module is implemented to establish a reference position at the start of operation. This ensures repeatability and accurate alignment of the sorting bins during each cycle.

Once the resistor value is identified, the ESP32 processes the data and rotates the stepper motor to align with the appropriate storage compartment. A sorting mechanism consisting of three servo motors is used to direct the resistor into the correct bin. These servo motors perform actions such as guiding, pushing, or dropping the resistor into designated sections based on its value range. The coordinated operation of the stepper motor and servo motors ensures efficient and reliable sorting.

In addition to sorting, the system also supports retrieval functionality. The stepper motor can rotate the storage unit to a desired position, allowing users to easily access specific resistors when required. This feature significantly reduces the time spent searching for components and enhances overall usability.

The integration of precise resistance measurement, accurate motor control, and intelligent processing makes the system highly reliable and efficient. By automating the sorting and retrieval process, the system minimizes human error, improves organization, and enhances productivity. Overall, the Resistor Sorter and Retriever System provides a practical and cost-effective solution for automated electronic component management.